

Robot Learning: From Fundamentals to Foundation Models

Lecture 3: Imitation Learning

Oier Mees

ETH zürich



Microsoft

02.03.2026

Feedback → Actions

- Moved from 3 papers to 2 papers per lecture
- Extra paper slots on Thursdays in IFW A 32.1
- HW 2 released, due on March 12th
- We can see when you use AI in autograder

Robot Learning: From Fundamentals to Foundation Models

Lecture 3: Imitation Learning

Oier Mees

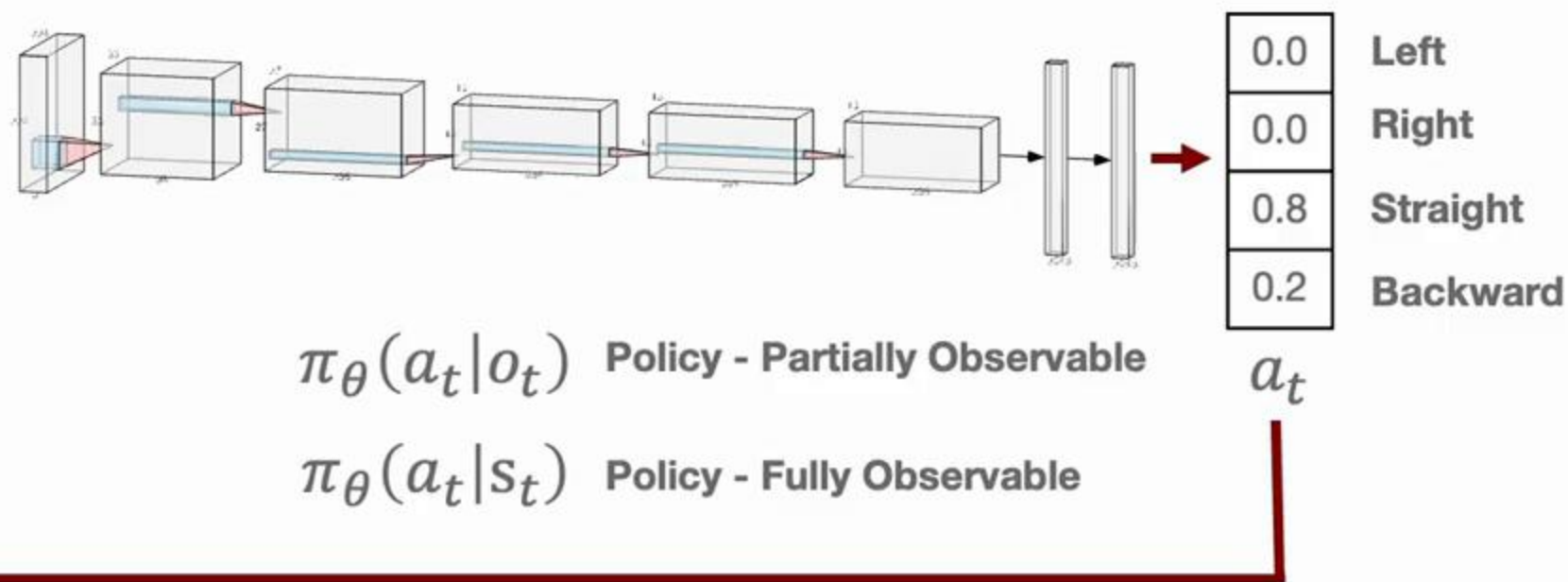
ETH zürich



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Recap: Mapping Observations to Actions



a_t - action: the decision taken by the agent at time t

o_t - observation: what the agent observes at time t

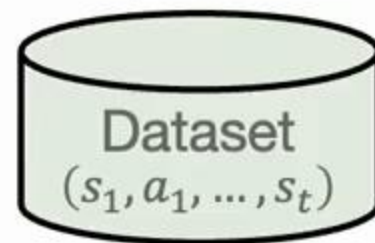
s_t - state: the “state” of the world at time t

τ trajectory: sequence of state and actions $(s_1, a_1, \dots, s_T, a_T)$

$r(s, a)$ reward: how good is s, a ?

Imitation Learning

- **Data:** Given trajectories collected by an “expert” “demonstrations” $\mathcal{D} = \{(s_1, a_1, \dots, s_t)\}$



- **Goal:** Learn a policy π_θ that “imitates” the expert’s behavior



Behavioral Cloning

1. Given expert demonstrations: $\mathcal{D} = \{(s_1, a_1, \dots, s_t)\}$
2. For deterministic policy, regress to expert's actions:
$$\min_{\theta} \frac{1}{|\mathcal{D}|} \sum_{(s,a) \in \mathcal{D}} ||\mathbf{a} - \hat{\mathbf{a}}||^2 \text{ where } \hat{\mathbf{a}} = \pi_{\theta}(\mathbf{s})$$
3. Deploy policy π_{θ} on robot

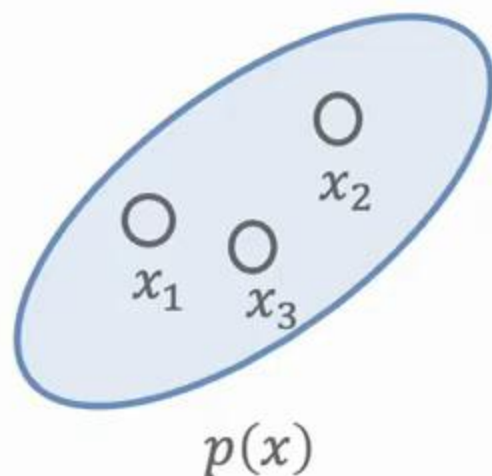
Learning to Fly in Swiss Forests

- Same camera trick



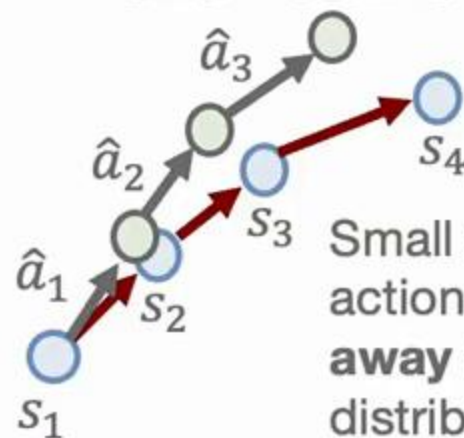
What can go wrong?

Supervised Learning



Inputs are independently, identically distributed (i.i.d.) and **independent of predicted labels!**

Behavioral Cloning



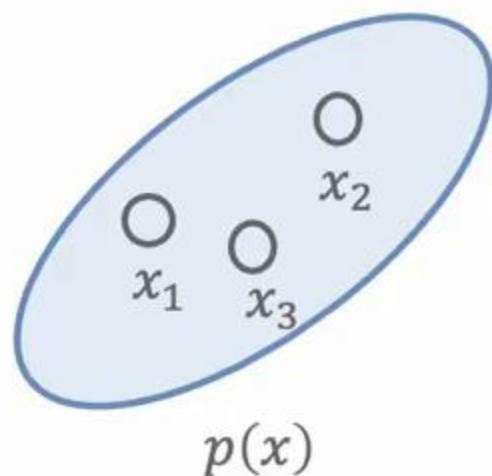
Small error in predicted action can lead to **drift away** from training data distribution!

$$p_{expert} \neq p_{\pi}(s)$$

States visited by expert States visited by the policy

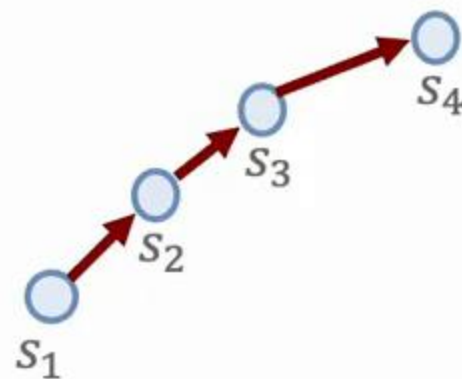
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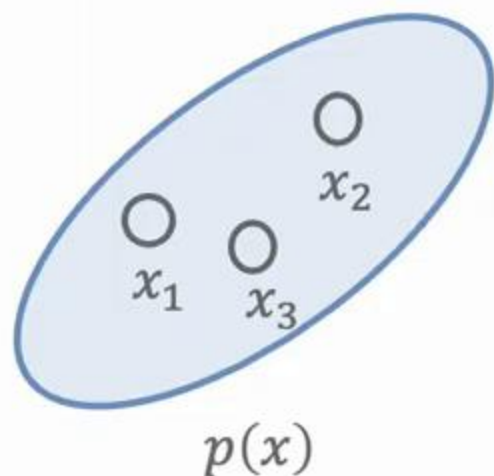
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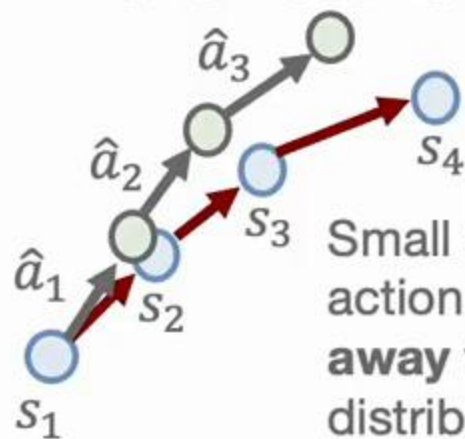
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Behavioral Cloning



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States visited by expert States visited by the policy

Does it work?



End to End Learning for Self-Driving Cars
Bojarski, et. al., 2016

Does it work?



It began its driving lessons in a parking lot

End to End Learning for Self-Driving Cars
Bojarski, et. al., 2016

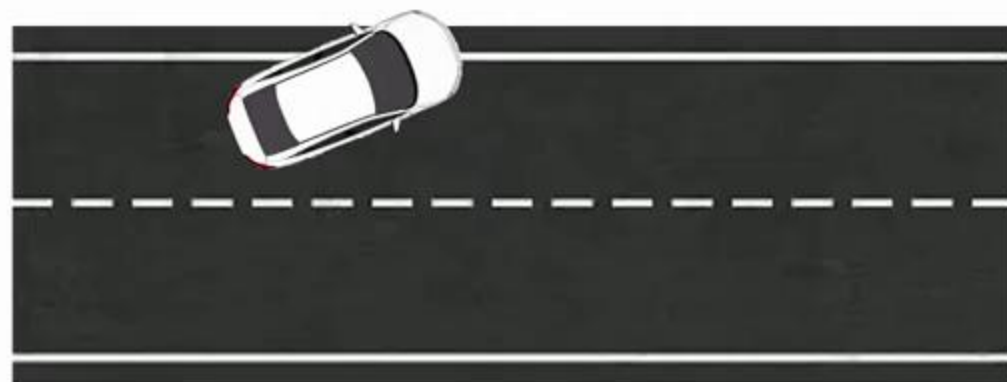
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End to End Learning for Self-Driving Cars
Bojarski, et. al., 2016

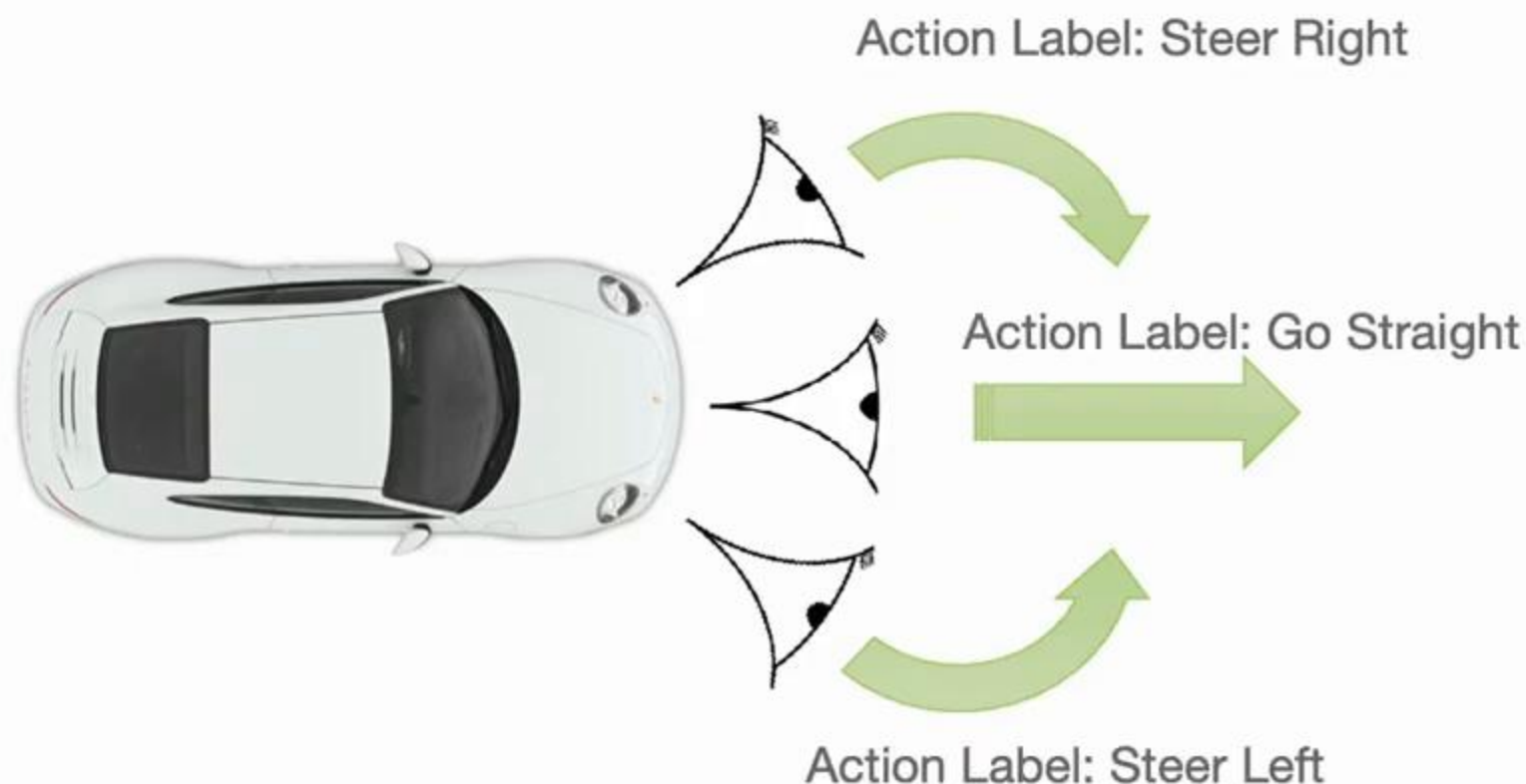
Why did it work?

What should the policy do here?
Expert never “visited this state”!



Why did it work? Data Augmentation!

- Add “fake” data that illustrates corrections with side-facing cameras $\mathcal{D}_{\text{aug}} = \{(\mathbf{o}_{\text{center}}, \mathbf{a}), (\mathbf{o}_{\text{left}}, \mathbf{a} + \delta), (\mathbf{o}_{\text{right}}, \mathbf{a} - \delta)\}$



Learning to Fly in Swiss Forests

- Same camera trick



Learning to Fly in Swiss Forests

- Same camera trick



Learning to Fly in Swiss Forests

- Same camera trick

 trail heading left	 trail straight ahead	 trail heading right
		
		

which direction is the trail heading to?

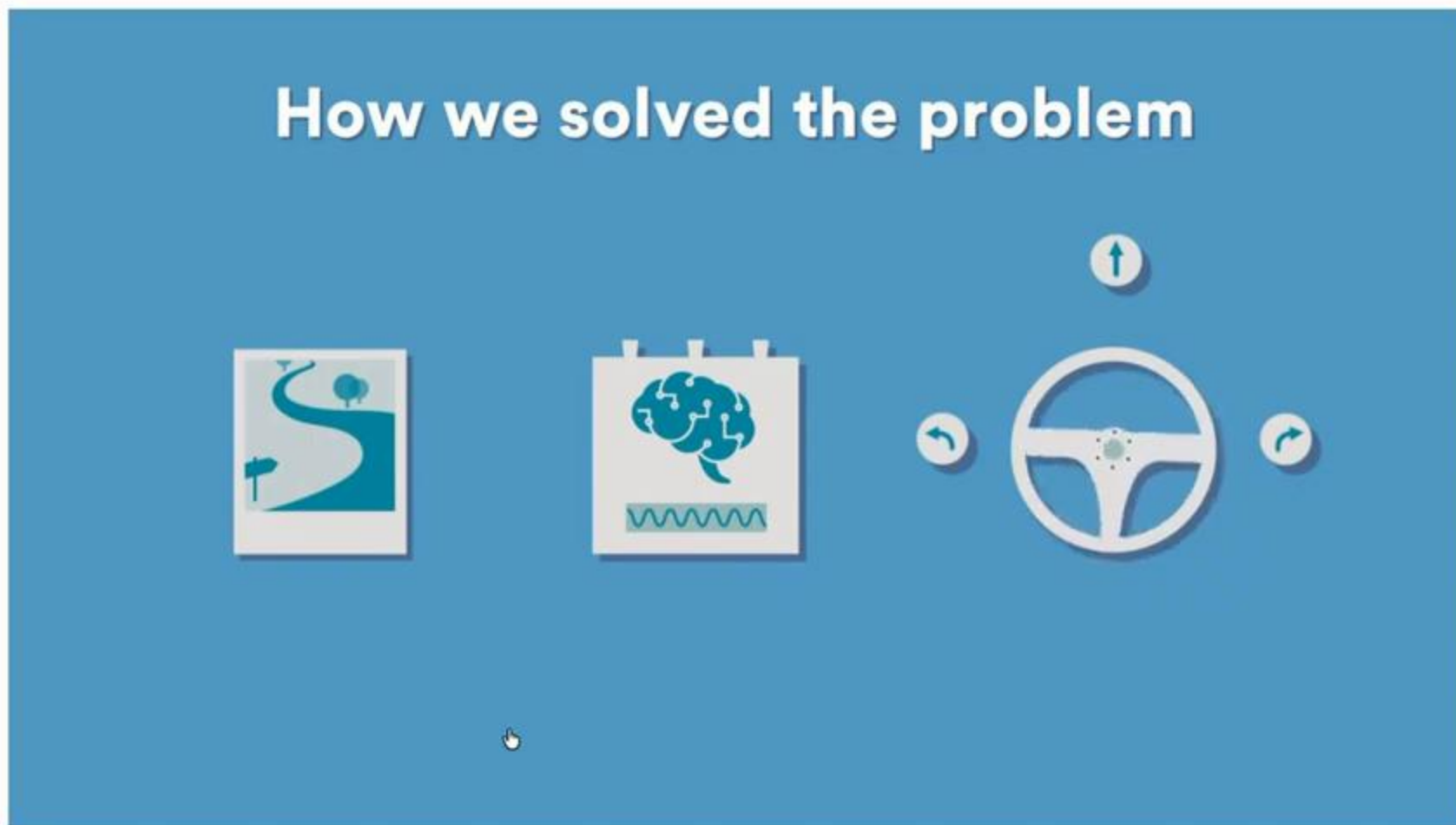
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How we solved the problem

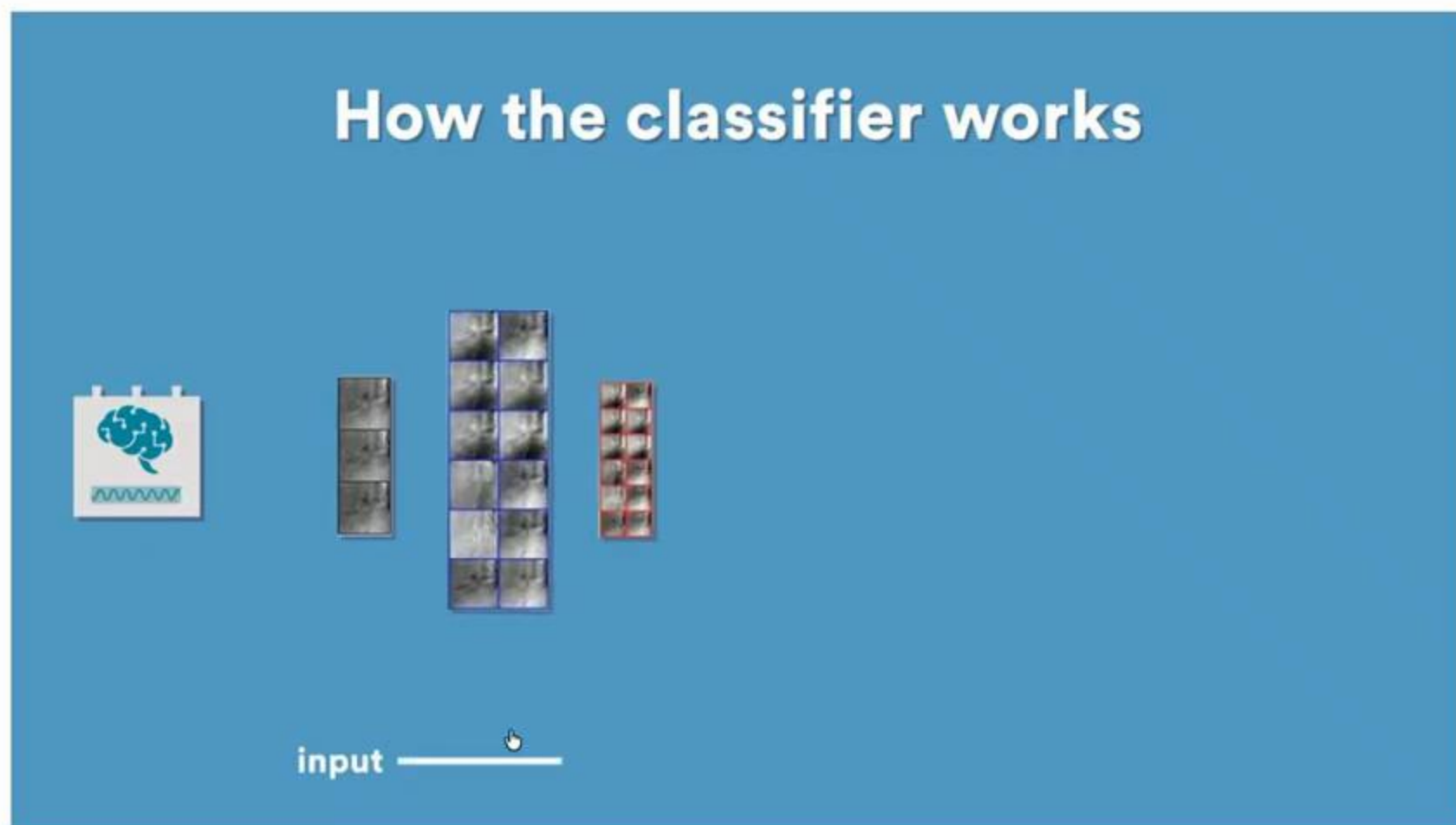
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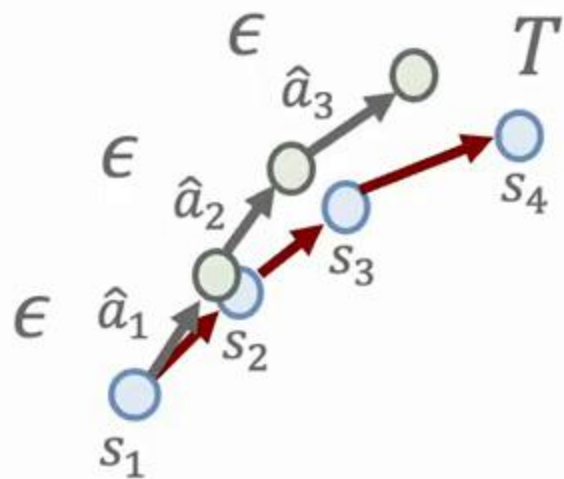


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Upper Bound of Behavioral Cloning



T : time horizon

ϵ : probability of π_θ making a mistake at any steps

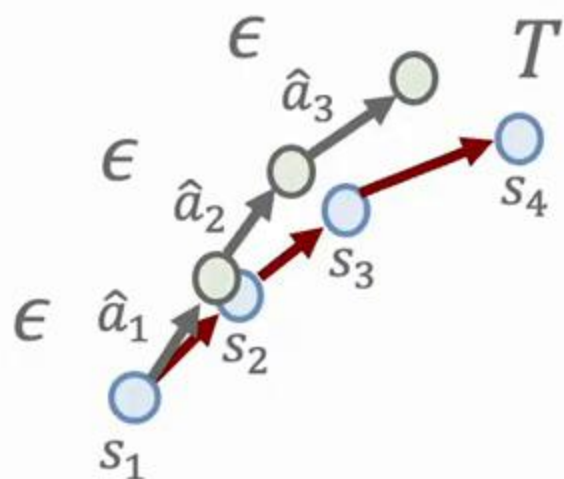
$$c(\mathbf{s}, \mathbf{a}) = \begin{cases} 0 & \text{if } \mathbf{a} = \pi^*(\mathbf{s}) \\ 1 & \text{otherwise} \end{cases}$$

$$P(\mathbf{a} \neq \pi^*(\mathbf{s}) \mid \mathbf{s}) \leq \epsilon, \quad \forall \mathbf{s} \in \mathcal{D}_{\text{train}}$$



More analysis: "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning" by Ross et al., 2011.

Upper Bound of Behavioral Cloning



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$$\mathbb{E} \left[\sum_t c(s_t, a_t) \right] \leq \epsilon T + (1 - \epsilon)(\epsilon(T - 1) + \dots) \approx \mathbf{O}(\epsilon T^2)$$



Distribution shift causes the error to grow **quadratically**!

More analysis: "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning" by Ross et al., 2011.

Addressing Compounding Error

Can we make?

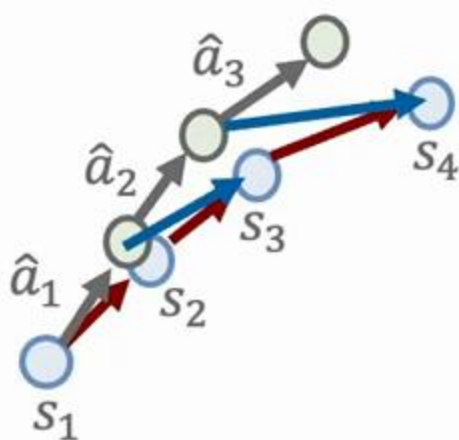
$$p_{expert} = p_{\pi}(s)$$

States visited by expert States visited by the policy

Aggregating Corrective Behavior

Dagger: Dataset Aggregation

1. Roll out policy π_θ and collect states: $(s'_1, \hat{a}_1, \dots, s'_t)$
2. Query/label expert action at visited states $\mathbf{a}^* \sim \pi_{\text{expert}}(\cdot | \mathbf{s}')$
3. **Aggregate** corrections with existing data $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s', \mathbf{a}^*)\}$
4. Update policy $\theta \leftarrow \arg \min_{\theta} \mathcal{L}(\pi_\theta, \mathcal{D})$



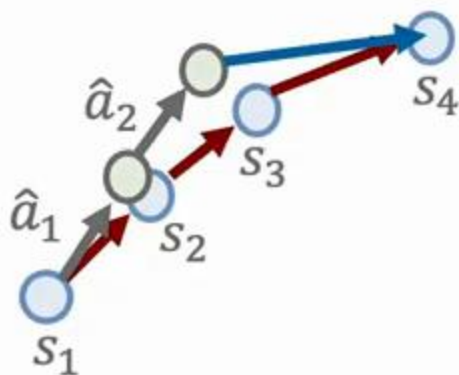
- ✓ Algorithm will converge $p_\pi(s) = p_{\text{expert}}$
- ✓ Achieves $O(\epsilon T)$ instead of BC $O(\epsilon T^2)$
- ✗ Hindsight labeling & expert querying difficult

Paradox BC works better if the data has more mistakes and recoveries!

Aggregating Online Interventions

Human Gated Dagger

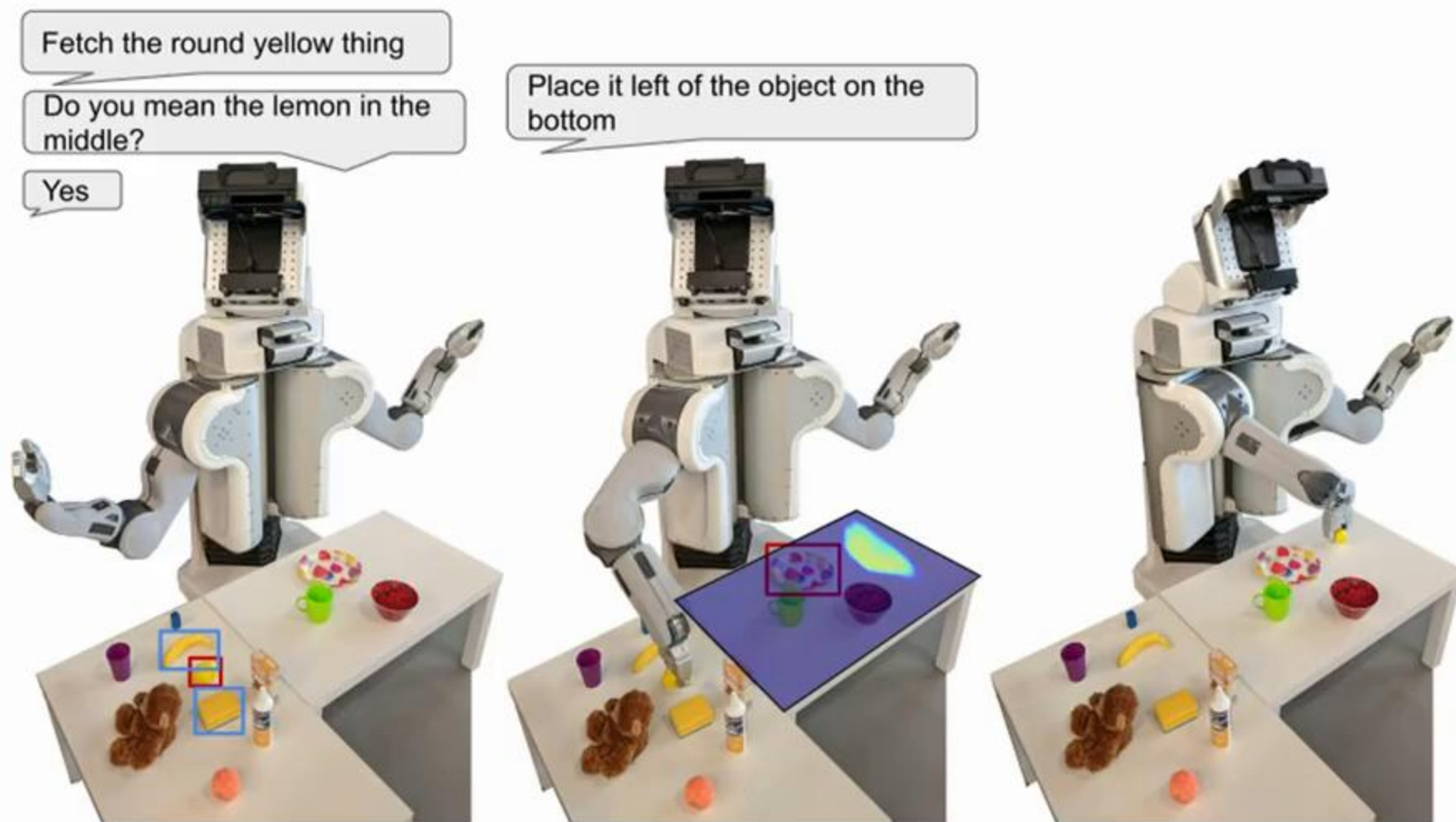
1. Roll out policy π_θ and collect states: $(s'_1, \hat{a}_1, \dots, s'_t)$
2. Expert intervenes at time t when policy makes mistake
3. Expert provides partial demo $(s'_t, \mathbf{a}_t^*, \dots, s'_T)$
4. **Aggregate** new demos $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s'_i, \mathbf{a}_i^*)\}; i \geq t$
5. Update policy $\pi_\theta \leftarrow \min_{\theta} \mathcal{L}(\pi_\theta, \mathcal{D})$



✓ Lets the human take control!

✗ How to detect when an intervention is needed?

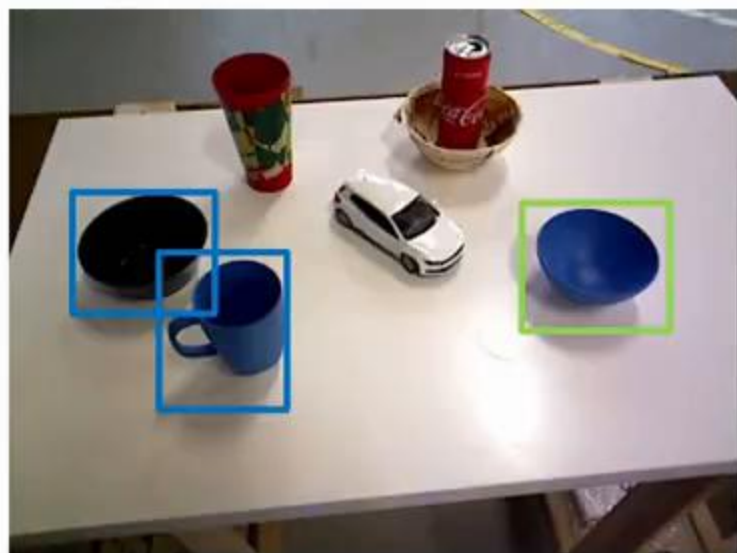
Case Study: Language Ambiguities



Composing Pick-and-Place Tasks By Grounding Language
Mees and Burgard, ISER 2020

Case Study: Language Ambiguities

"Pick up the blue stuff"

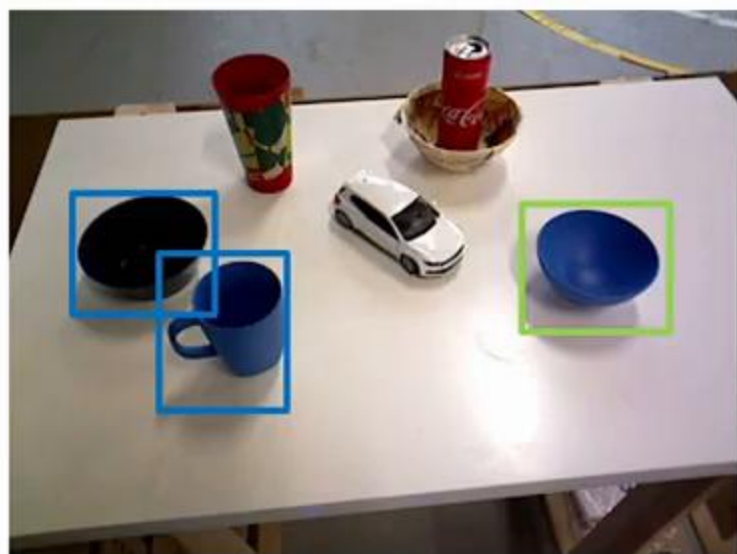


Do you mean this
blue bowl on the table?



Case Study: Language Ambiguities

“Pick up the blue stuff”



Do you mean this
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1. Train referential expression comprehension model with ranking loss

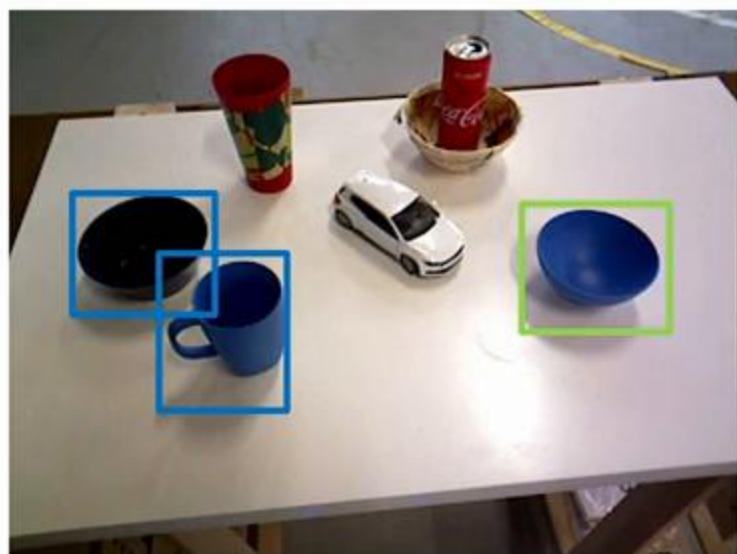
$$\mathcal{L}_{\text{rank}} = \sum_i \left[\lambda_1 \max(0, m_1 + S(o_i | r_j) - S(o_i | r_i)) + \lambda_2 \max(0, m_1 + S(o_k | r_i) - S(o_i | r_i)) \right]$$

2. Train referential expression generation model

$$\mathcal{L}_{\text{gen}} = - \sum_i \log P(r_i | v_i)$$
$$\mathcal{L}_{\text{mmi}} = \sum_i [\lambda_3 \max(0, m_2 + \log P(r_i | v_k) - \log P(r_i | v_i))]$$

Case Study: Language Ambiguities

"Pick up the blue stuff"



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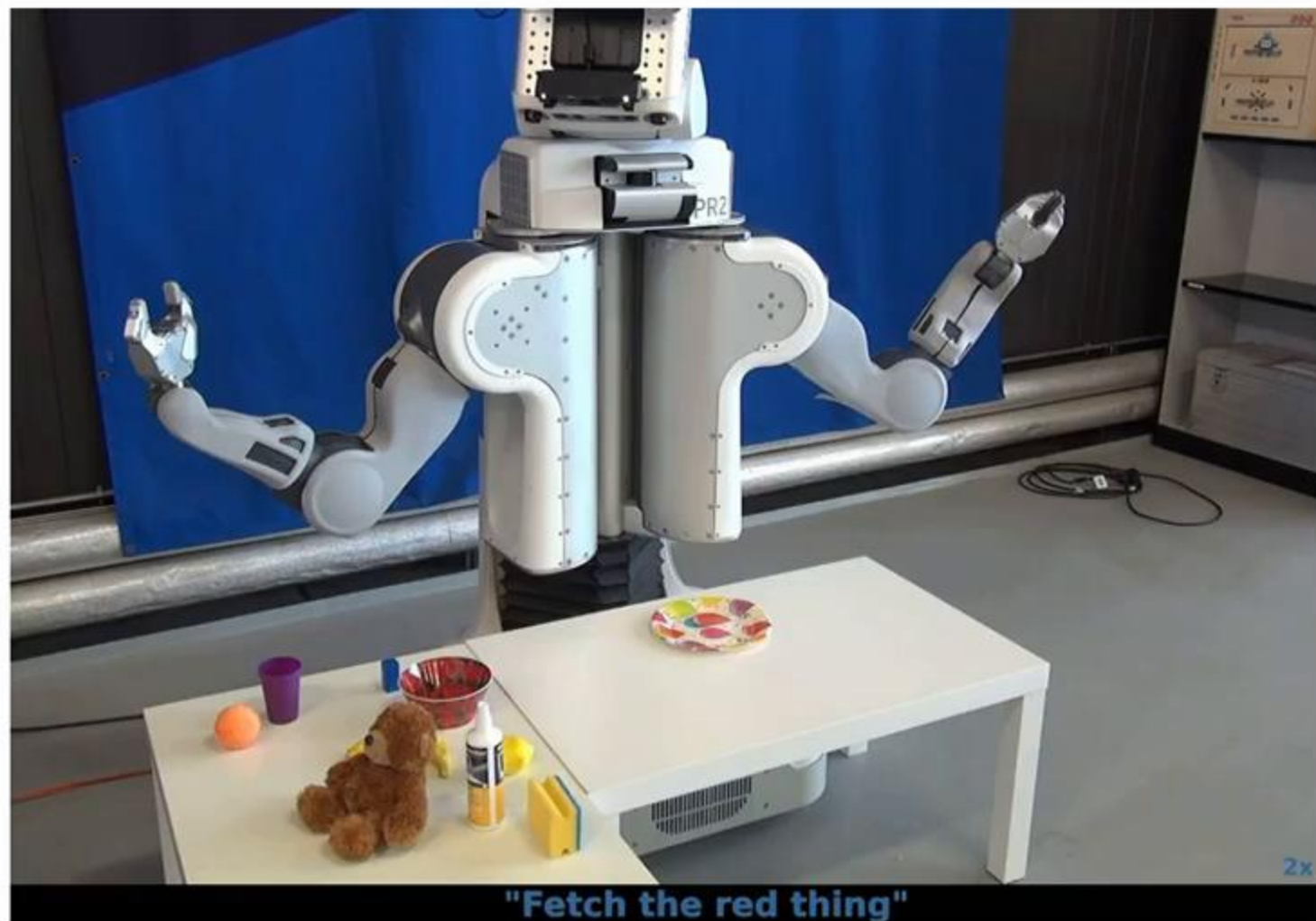
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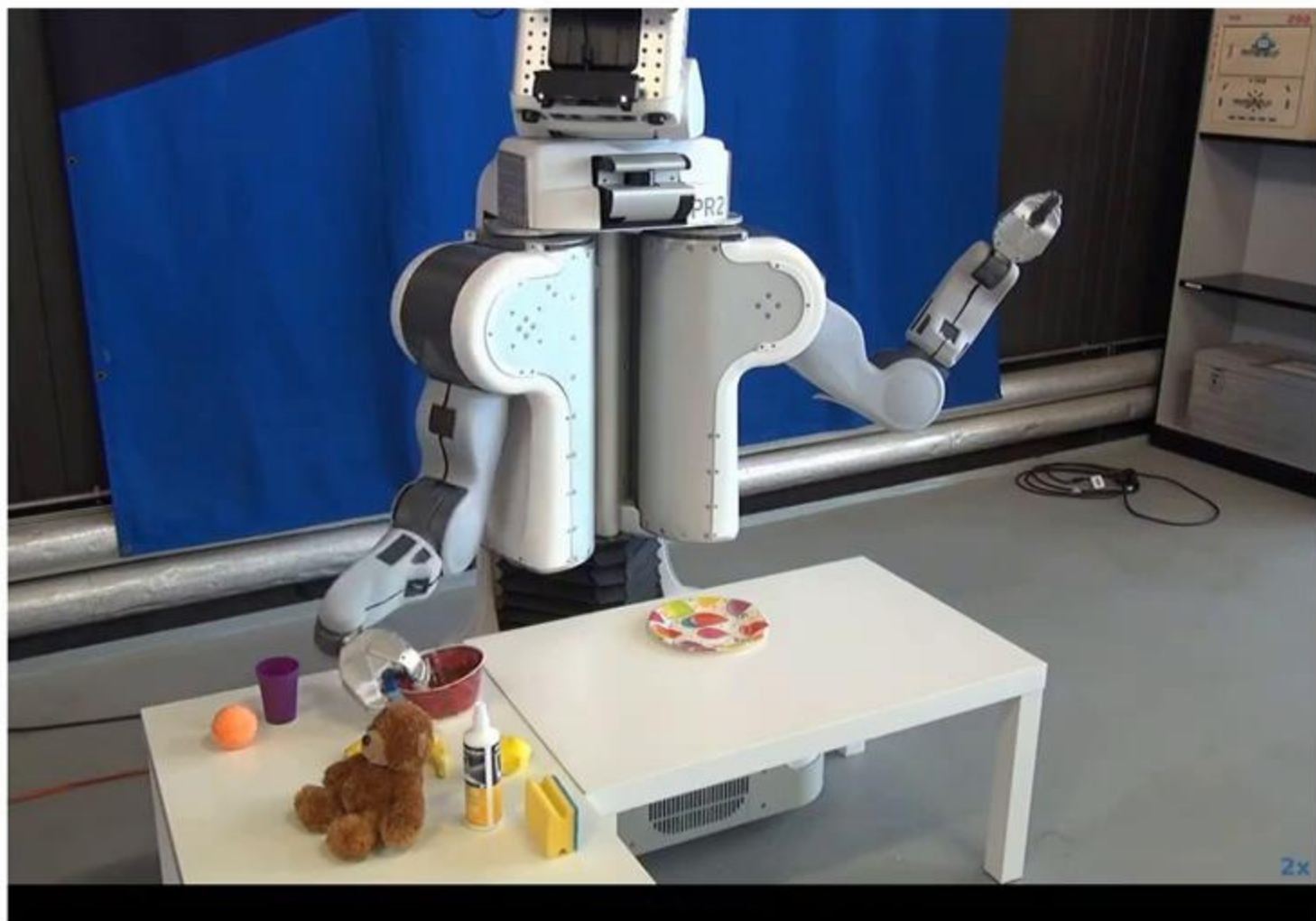
3. If at test time there are multiple objects within margin m , generate descriptions and ask user

Case Study: Language Ambiguities



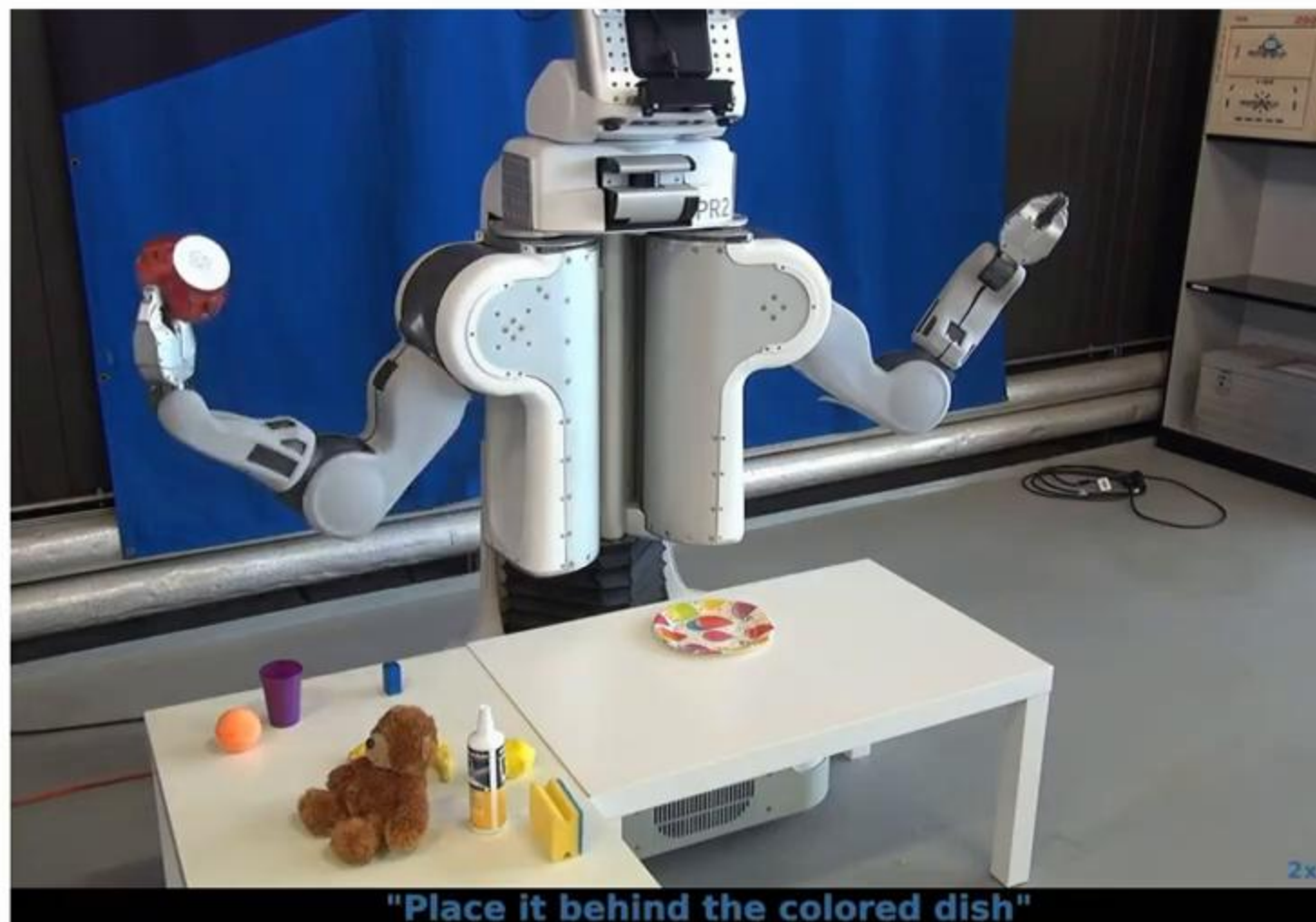
Composing Pick-and-Place Tasks By Grounding Language
Mees and Burgard, ISER 2020

Case Study: Language Ambiguities



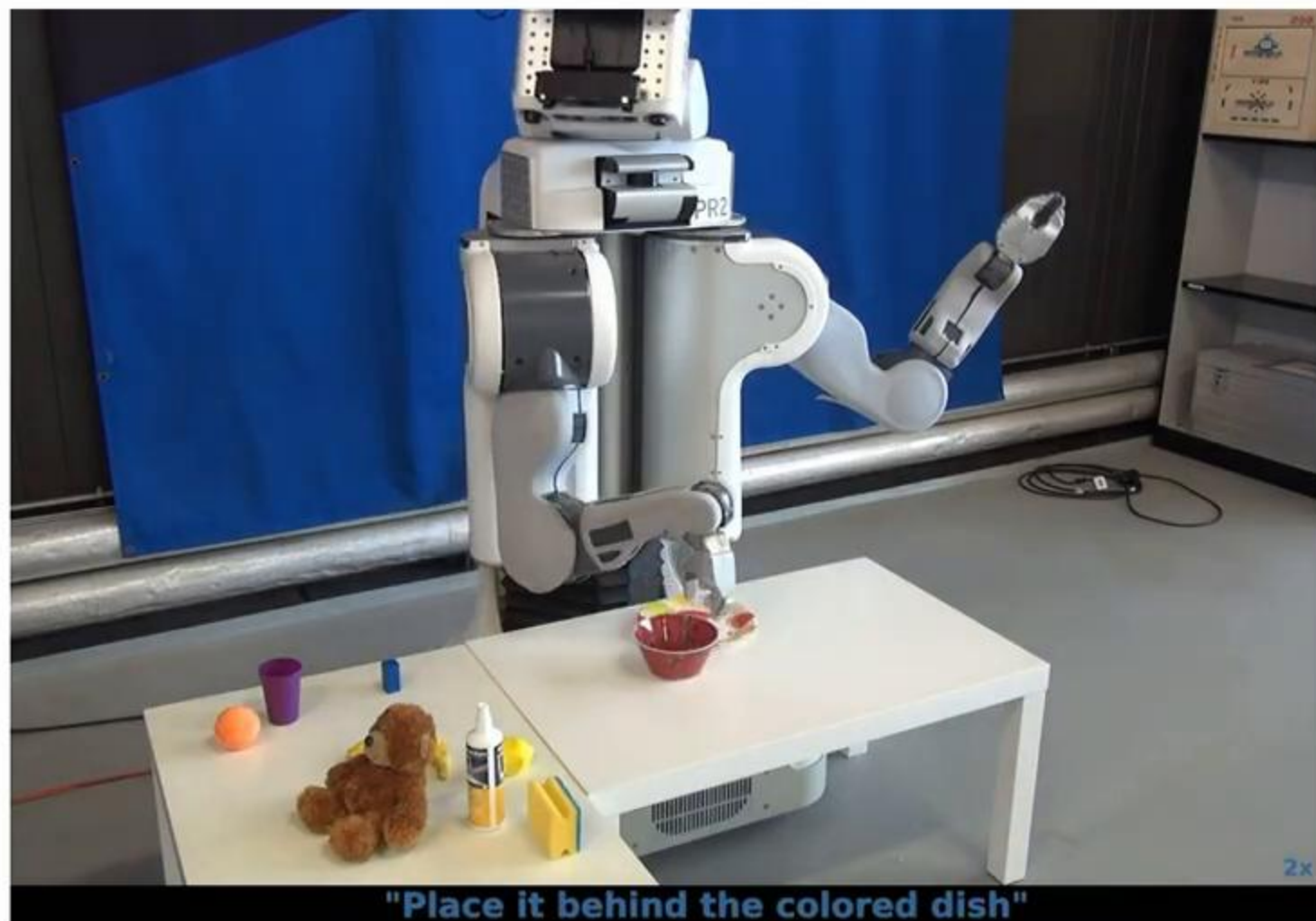
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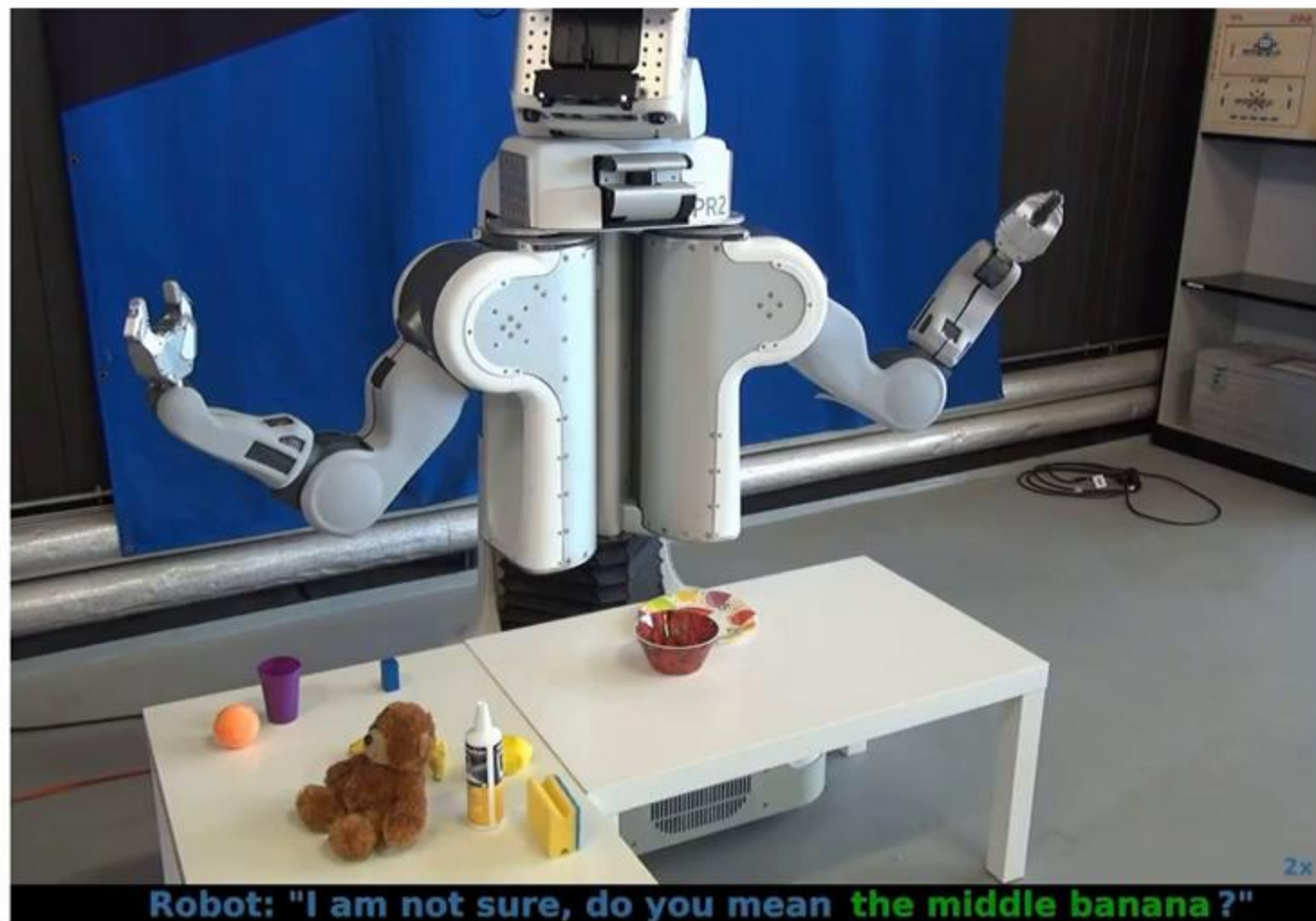
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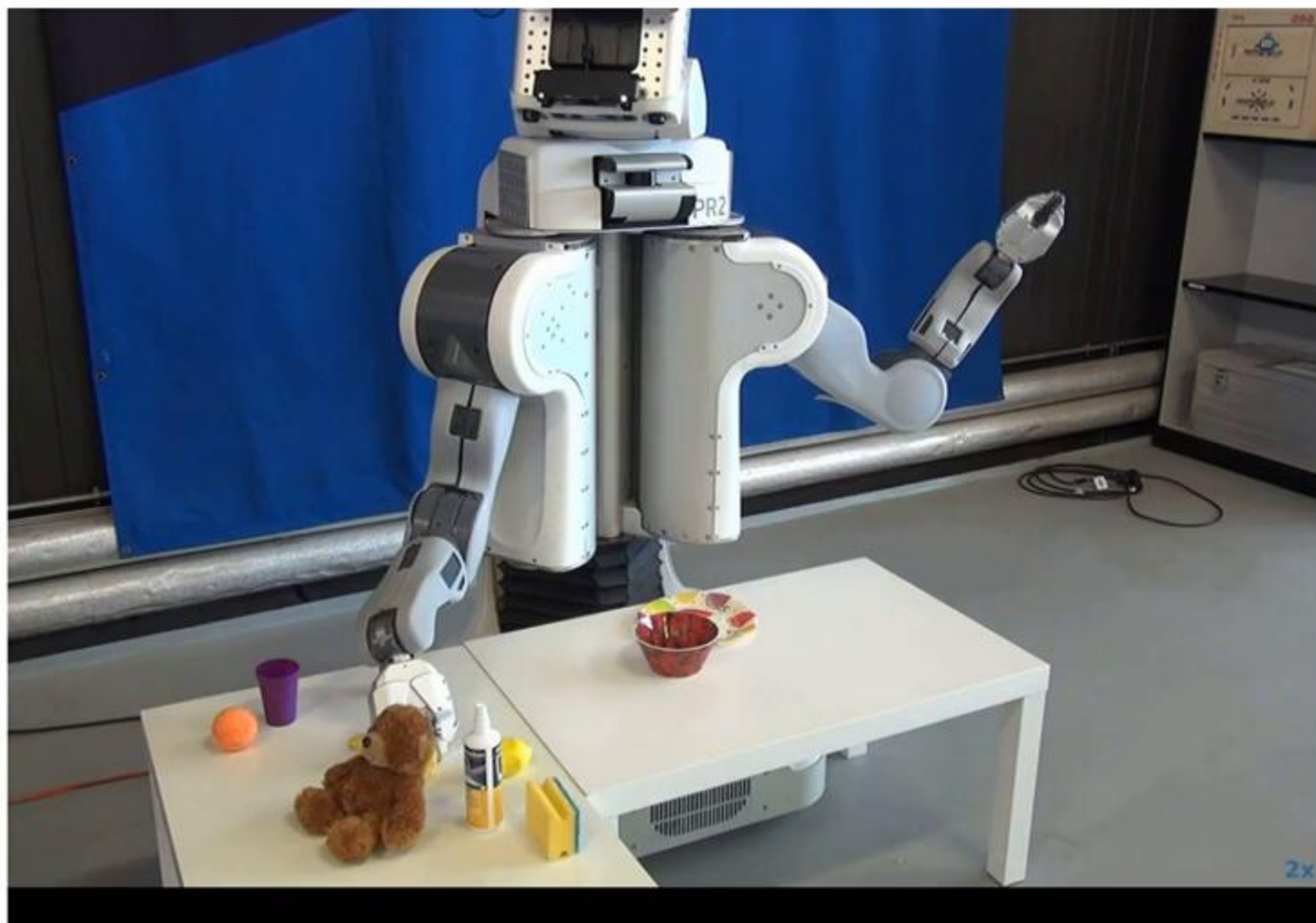
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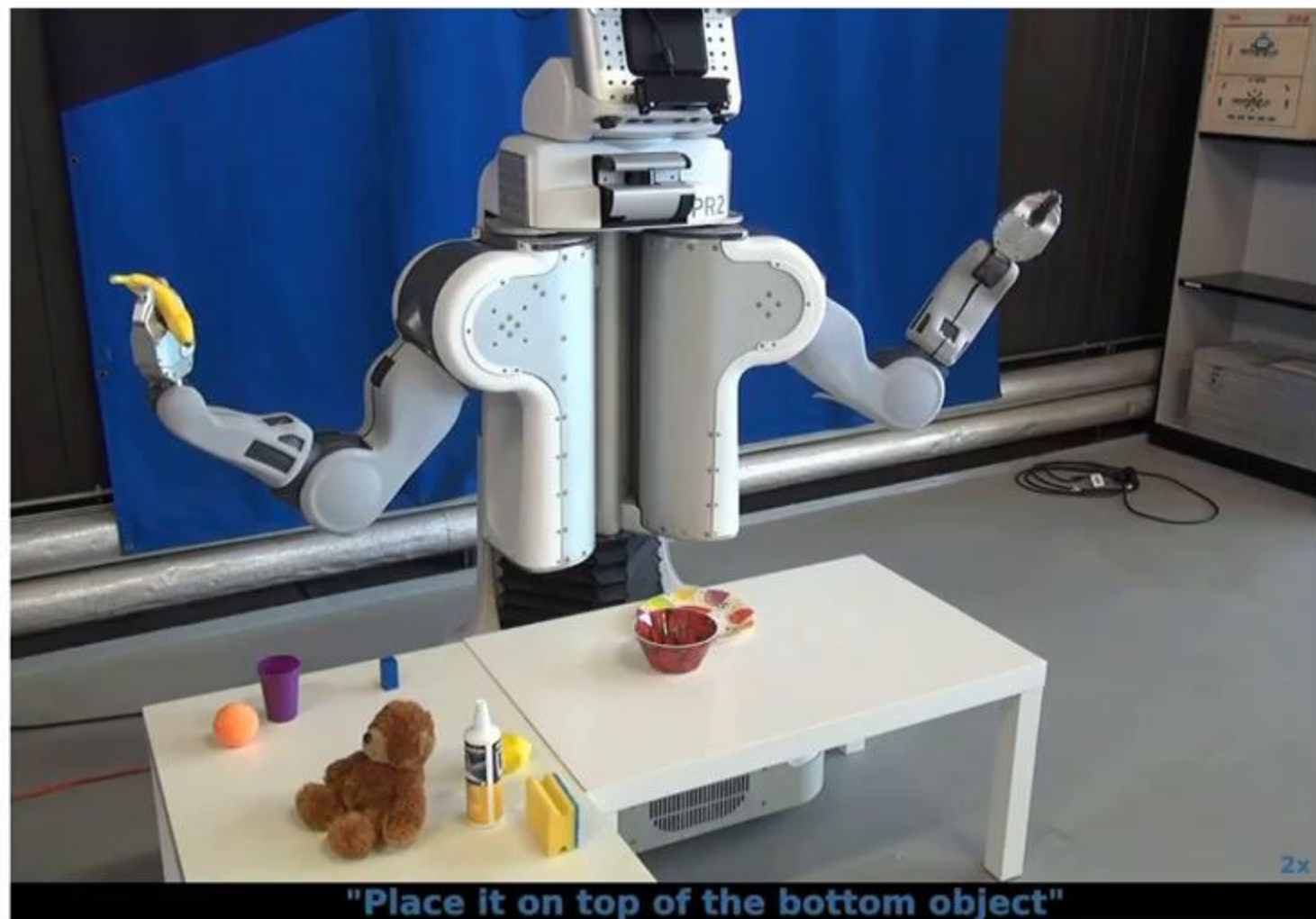
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Composing Pick-and-Place Tasks By Grounding Language
Mees and Burgard, ISER 2020

Why might we still fail to mimick the expert?

Non-Markovian behavior

$$\pi_{\theta}(a_t|o_t)$$



action depends only on
current observation

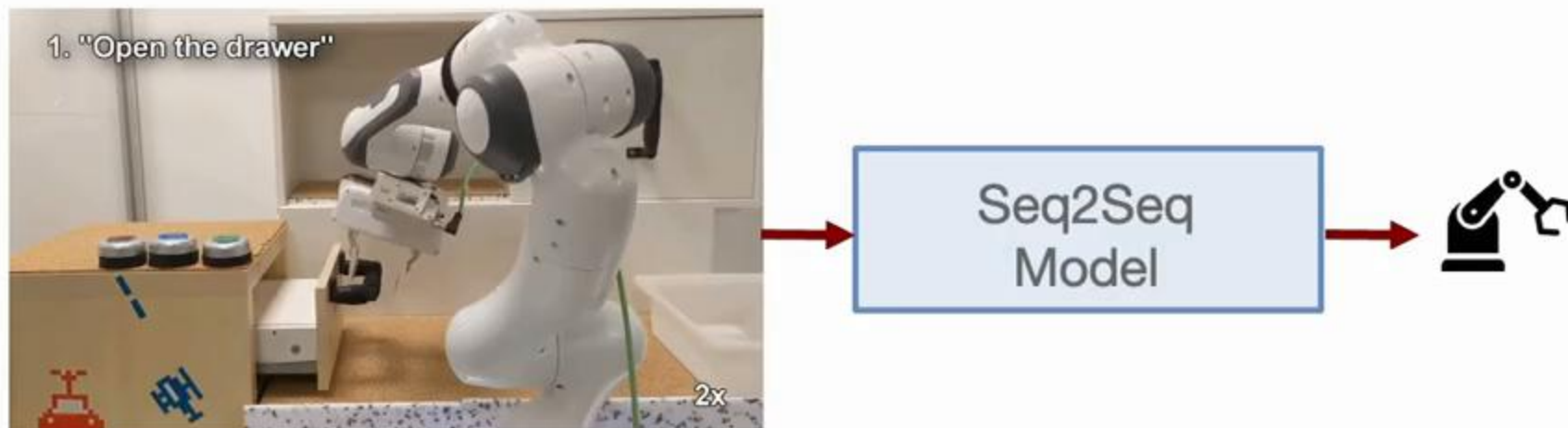
$$\pi_{\theta}(a_t|o_1, \dots, o_t)$$



Human behavior might be affected
by past observations, emotions,
privileged information etc.

Adding History to Policy

- Encode with some sequence to sequence architecture



✗ Adding history to the policy **doesn't always make things better!**

Causal Confusion

- Policy might infer **spurious correlations!**



1. Data: every time the robot opens the drawer, the gripper grasps the handle with 10 N
2. Model learns: if history shows a gripper force spike of 10 N, pull to open
3. Test time: gripper slips and only reads 2 N, the pull command is never triggered

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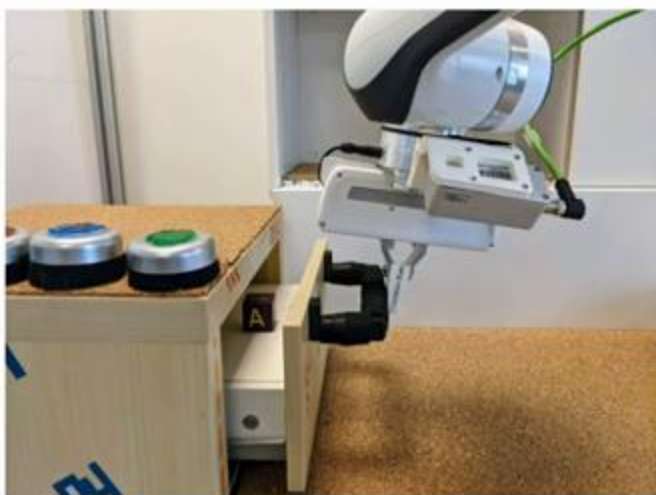
✗ Policy thinks history of gripper sensor causes open drawer instead of the visual state!

Why might we still fail to mimick the expert?

Multimodal behavior

Stochasticity: Many (**infinite**) ways to close a drawer with a 7-DoF robot!

Expert Inconsistency: Humans operators might use different '**modes**' across trials



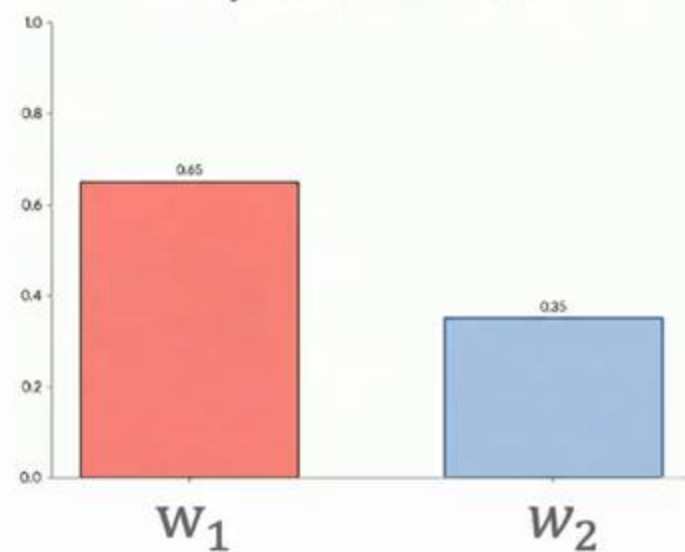
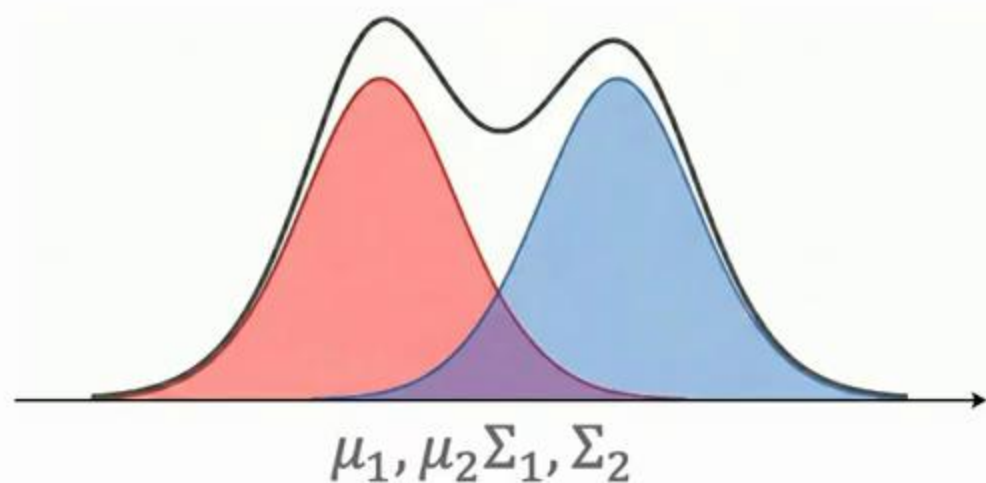
Multimodal Behavior

- Deterministic MSE BC policy will fail due to “**averaging**” of **modes**!



Mixture of Gaussian Distributions

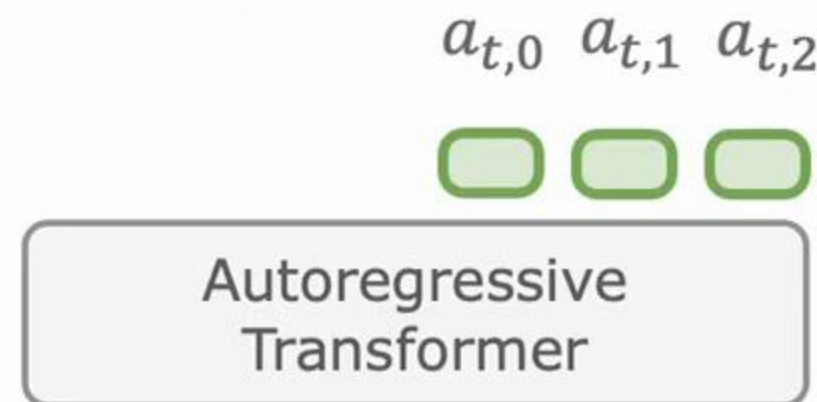
- More expressive than a single gaussian
- Need to predefine number of gaussians



$$\pi(\mathbf{a} | \mathbf{o}) = \sum_i w_i \mathcal{N}(\mu_i, \Sigma_i)$$

Autoregressive Discretization

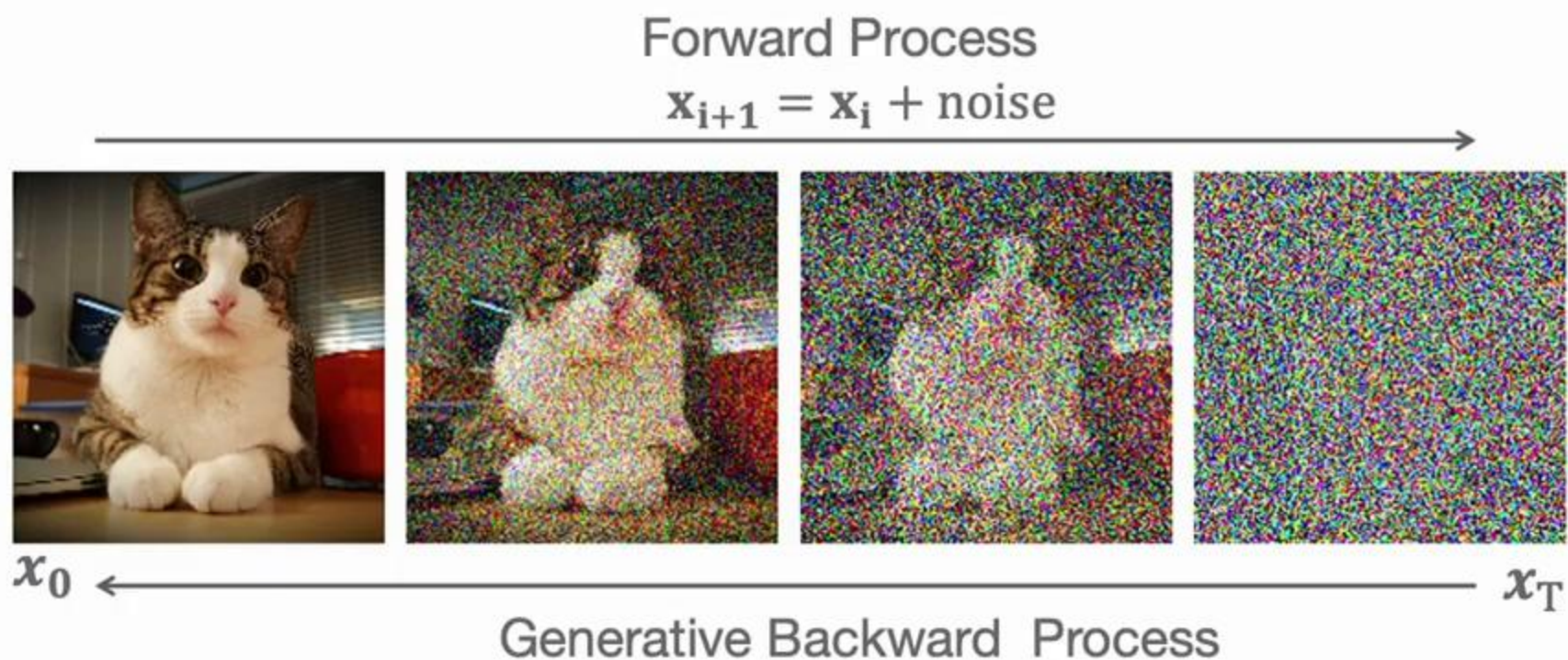
- Discretization great for representing multimodal distributions
- Impractical to scale to higher dimensions (exponential)
- Solution: **per-dimension discretization with sequence model**



$$p(\mathbf{a}_t | \mathbf{s}_t) = p(a_{t,0}, a_{t,1}, a_{t,2} | \mathbf{s}_t) = p(a_{t,2} | \mathbf{s}_t, a_{t,0}, a_{t,1}) p(a_{t,1} | \mathbf{s}_t, a_{t,0}) p(a_{t,0} | \mathbf{s}_t)$$

Diffusion

- Models complex distributions over continuous variables
- We can replace images with robot actions, more on W6



learn $f(x_i) = x_{i-1}$ in practice $f(\mathbf{x}_i) = \text{noise}$, $\mathbf{x}_{i-1} = \mathbf{x}_i - f(\mathbf{x}_i)$

Latent Variable Models

- Output still gaussian, but receives additional input
- Can represent “any” distribution
- Popular: conditional VAEs

$$\pi(\mathbf{a} \mid \mathbf{o}, \mathbf{z}) = f_{\theta}(\mathbf{o}, \mathbf{z})$$

Conditional Decoder



$z_{\text{left}} \sim \mathcal{N}(0, \mathbf{I})$

$z_{\text{right}} \sim \mathcal{N}(0, \mathbf{I})$

Learning Multiple Tasks

Latent Variable Models are different from task conditioning

“one task”

$$\pi_{\theta}(a_t | s_t)$$

Single task
Behavior Cloning
(Pomerleau 1991)

“multiple tasks”

$$\pi_{\theta}(a_t | s_t, \text{task id})$$

Task conditioned
Behavior Cloning
(Rahmatizadeh 2018)



Learning Multiple Tasks

“Move the sliding door to the left”

$$\pi_{\theta}(a_t | s_t, \text{task id})$$



Did it succeed? Where is the threshold, 50%?

Learning Multiple Tasks

“Move the sliding door to the left” $\pi_{\theta}(a_t | s_t, \text{task id})$



Did it succeed? Where is the threshold, 50%?

How to Scale Control to “Any” Tasks?

Tasks are often **continuous**, not discrete

“one task”

$$\pi_{\theta}(a_t | s_t)$$

Single task
Behavior Cloning
(Pomerleau 1991)

“multiple tasks”

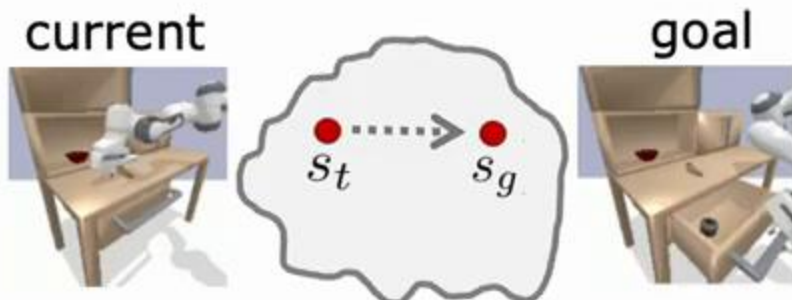
$$\pi_{\theta}(a_t | s_t, \text{task id})$$

Task conditioned
Behavior Cloning
(Rahmatizadeh 2018)

“any tasks”

$$\pi_{\theta}(a_t | s_t, s_g)$$

Goal state conditioned
Behavior Cloning
(Lynch 2019)



Let's put everything together!

**How can we learn any task with a latent variable model
that handles multimodality?**

Case Study: Play Data

- No upfront tasks
- Scalable, reset-free data collection
- Rich and highly multimodal behaviors



Case Study: Learning Policies Play Data

Sequence

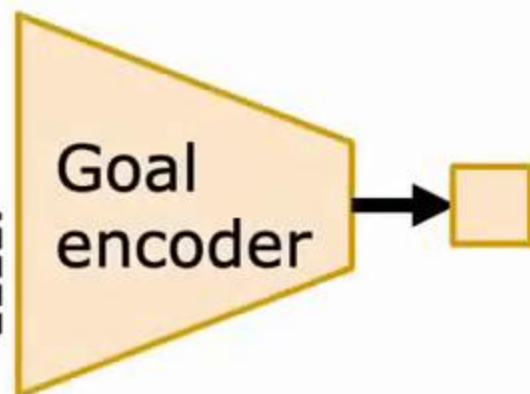


Case Study: Learning Policies Play Data

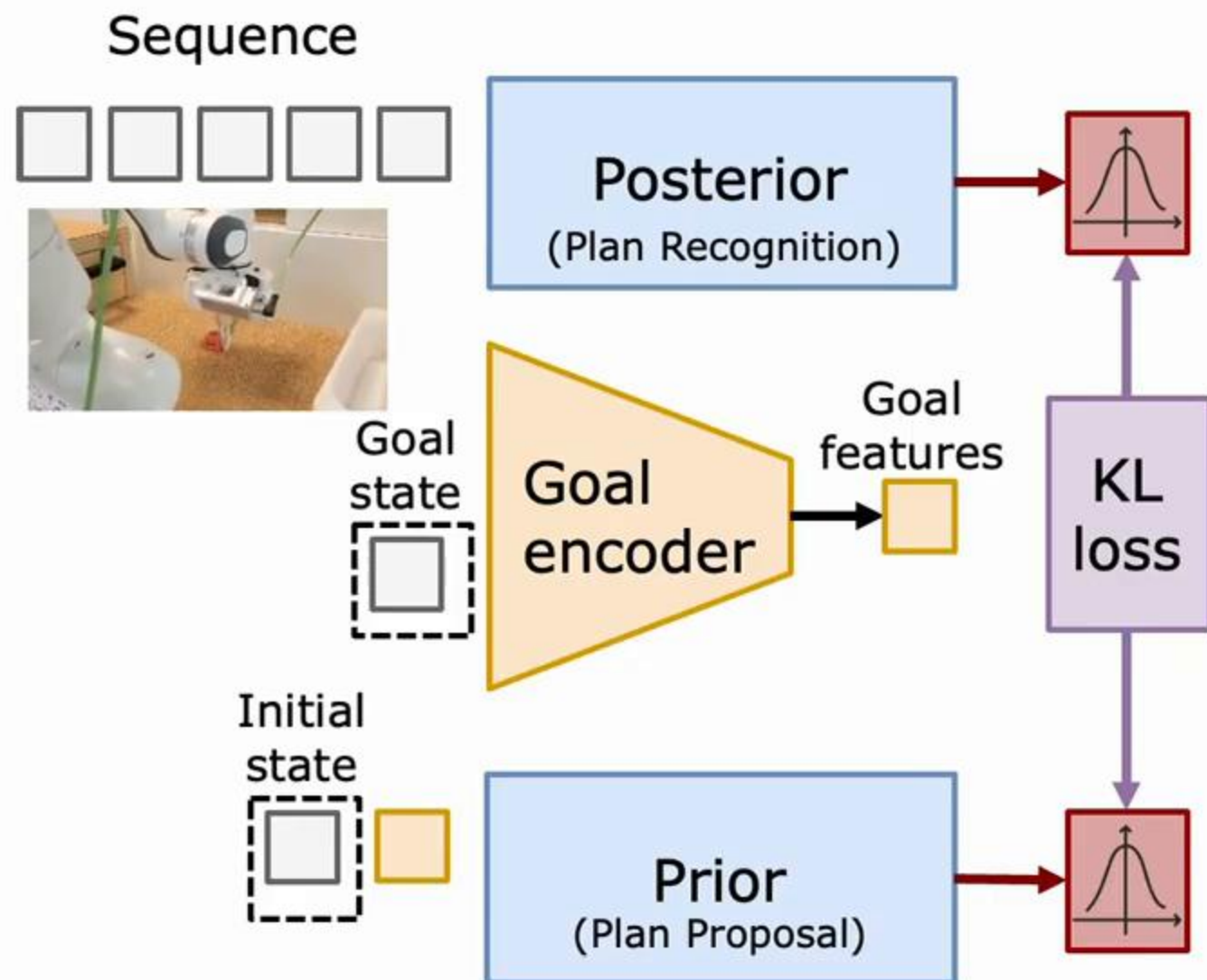
Sequence



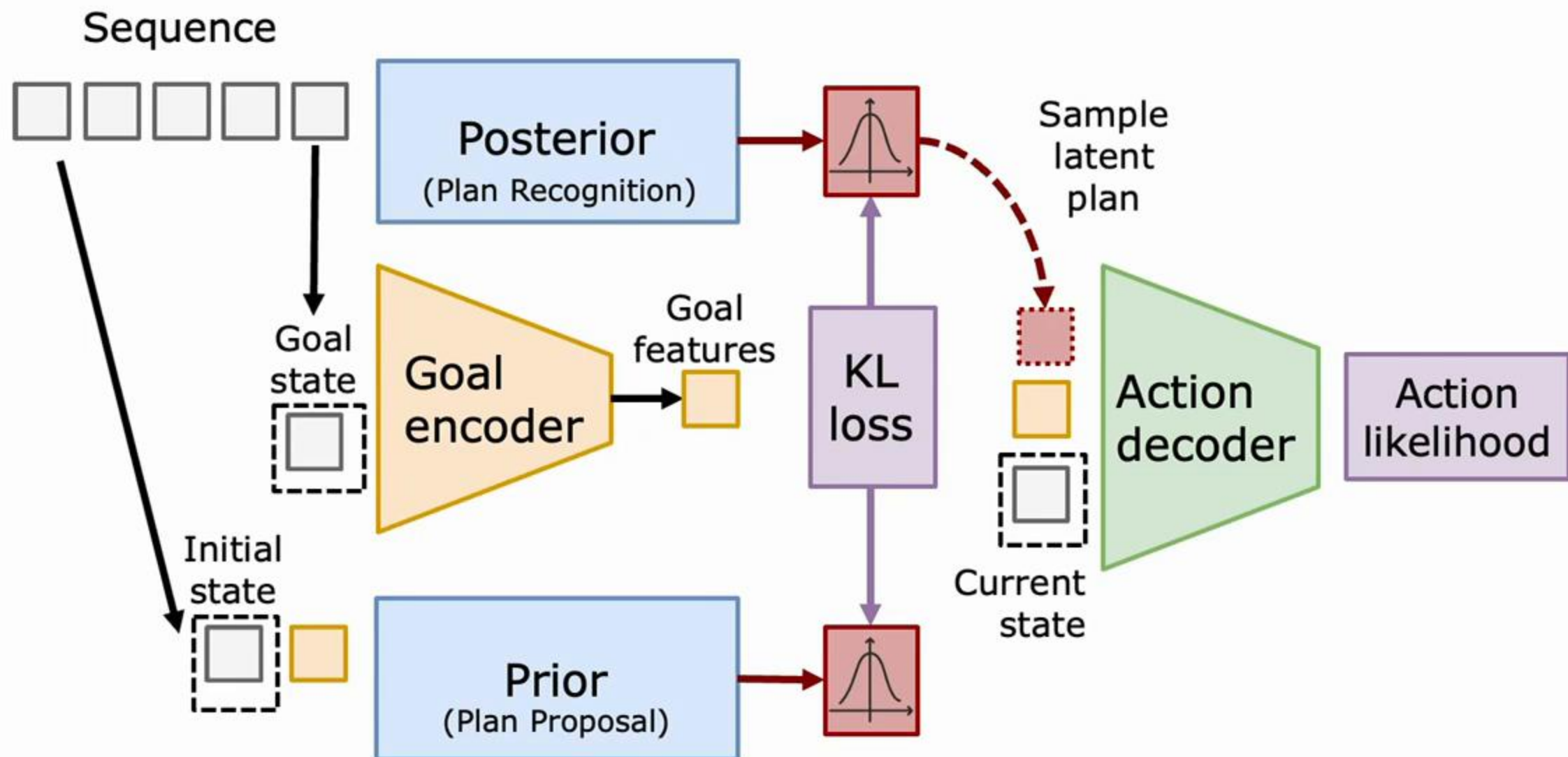
Goal
state



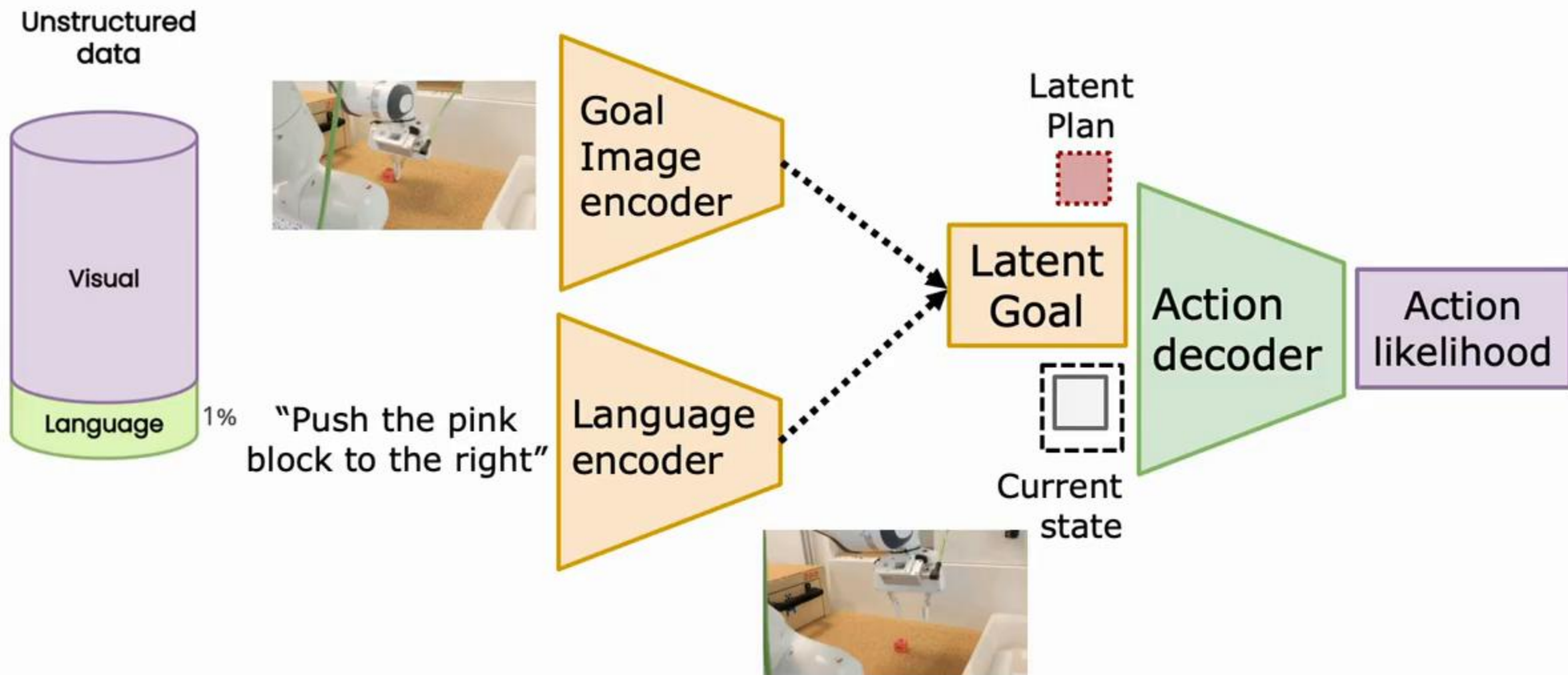
Case Study: Learning Policies from Play Data



Case Study: Learning Policies from Play Data



Case Study: Learning Policies from Play Data



The Math Behind Goal-Conditioned Latent Variable Models

trajectories; latent plans



$p_{\theta}(\tau, z)$  when z continuous, marginalization becomes intractable

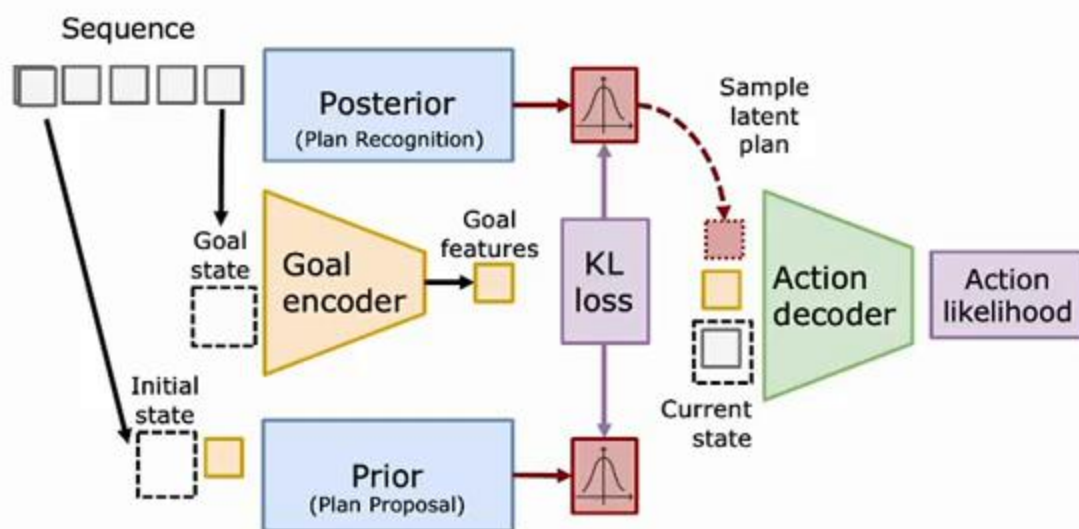
Optimize surrogate objective; variational lower bound of marginal log-likelihood

$$\log p_{\theta}(\tau) \geq \underbrace{-KL\left(q_{\phi}(z|\tau) \parallel p_{\theta}(z)\right)}_{\text{Regularization:}} + \underbrace{\mathbb{E}_{q_{\phi}(z|\tau)} [\log p_{\theta}(\tau|z)]}_{\text{Reconstruction:}}$$

Regularization:
are latent plan z close to our
prior distribution over τ ?

Reconstruction:
how well does latent plan z explain
the expert behavior τ ?

The Math Behind Goal-Conditioned Latent Variable Models



**Posterior –
Plan Recognition**

**Prior –
Plan Proposal**

Action Decoder

$$\log p_{\theta}(\tau|c) \geq -KL(q_{\phi}(z|\tau, c) || p_{\theta}(z|c)) + \mathbb{E}_{q_{\phi}(z|\tau, c)} [\log p_{\theta}(\tau|z, c)]$$

$$c \leftarrow (s_c, s_g)$$

My PhD: Natural Language + Learning from Play



CALVIN: A Benchmark for Language-conditioned Policy Learning for Long-horizon Robot Manipulation Tasks

Mees et. al., RA-L 2022

Best Paper Award RA-L 2022



What Matters in Language Conditioned Imitation Learning over Unstructured Data

Mees, et al. RA-L 2022.



Grounding Language with Visual Affordances over Unstructured Data
Mees, et al. ICRA 2023.

Finalist Best Paper Award ICRA 2023



Affordance Learning from Play for Sample-Efficient Policy Learning

Borja*, Mees* et. al., ICRA 2022



Latent Plans for Task Agnostic Offline Reinforcement Learning

Rosete*, Mees* et. al., CoRL 2022

Conclusion: Behavioral Cloning Failures

Distribution Shift

- Dagger
- Human Gated Dagger



Detect when to intervene

Non-Markovian Behavior

- Privileged Observation
- History & Causal Confusion

Multimodal Behavior

- Mixture of Gaussians
- Autoregressive Discretization
- Diffusion
- Latent Variable Models

Thank you for your attention